If one more lecture required then I will write an email along with the date

- Diffie Hellman key exchange.

E, $P \in E$, (aP, bP) for public

(but a & b are private)

$a \to$ Alice, $b \to$ Bob

Alice sends aP to Bob &

Bob sends bP to Alice

$$\left.\begin{array}{ll} \text{Alice finds} & a(bP) \\ \text{\& Bob} \quad " & b(aP) \end{array}\right\} \text{ in } E$$

Since E abelian $a(bP) = b(aP)$

is the common knowledge.

Massey-Omura Crypto system.

E is fixed (public)

A message $M \in E$ is to be sent.

~~Secret~~ private key a is with Alice

& " " b is with Bob.

Alice sends ~~to B~~ aM to Bob.

Bob ~~to~~ finds $b(aM)$ & sends $\underline{baM = abM}$ to Alice

Alice finds $a^{-1}(abM) = (a^{-1}a)bM = bM$

& ~~sends~~ sends bM to Bob.

Bob finds $b^{-1}(bM) = M$.

__Remark:__ Since a^{-1} & b^{-1} are required

so a & b (each) must be coprime to

$n = \#(E)$.

__E.g.,__ $E = \{(x,y) \in \mathbb{F}_{11}^2 \mid y^2 = x^3 + 3x + 1\} \cup \{O\}$

Recall $\#(E) = 18$

Choose a (by Alice) $= 7$ so that $7^{-1} = 13$

" b (by Bob) $= 11$ " " $11^{-1} = 5$

Suppose $M = \underbrace{(6,2)}$ $\in E$

$$\boxed{\begin{aligned} &6^3 + 3 \times 6 + 1 \\ &= 7 + 7 + 1 \\ &= 4 \\ &= 2^2 \end{aligned}}$$

Alice computes $7M = (3,2)$ sends it to Bob.

Bob " $11(7M) = 11(3,2) = (1,7)$ sends it to Alice

~~B~~ & Alice finds $7^{-1}(11(7M)) = 7^{-1}(1,7) = 13(1,7)$

$= (3,9)$

Sends it to Bob.

& Bob finds $M = 11^{-1}(11(7M)) = 5(3,9) = (6,2)$

Elgamal Cryptosystem on Elliptic curve.

An Elliptic curve E & a pt $P \in E$ are fixed.

Alice has the private key a, she computes aP & sends it to Bob.

Bob wants to send message $M (\in E)$ to Alice.

Bob picks $k \in \mathbb{N}$ & find
$k(aP) = kaP$. & kP

Bob finds $Q_2 = M + kaP \in E$

Bob sends $(Q_1, Q_2) \in E^2$

where $Q_1 = kP$

Alice knows that the message is the second component i.e. Q_2

She computes $aQ_1 = a(kP) = kaP$

& finds $Q_2 - aQ_1 = M + kaP - kaP$
$= M$

E.g., $E = \{(x,y) \in \mathbb{F}_{11}^2 \mid y^2 = x^3 + 5x + 1\} \cup \{\infty\}$

$P = (1,4)$ which are in public domain.

Alice chose $a = 11$ (which is secrete key to Alice)

She finds $aP = 11(1,4) = (6,9)$

Sends it to Bob.

Bob ~~finds~~ picks $k = 13$

& finds $k(aP) = 13(6,9)$
$= (1,7)$

Bob wants to send $M = (3,2)$

$Q_2 = M + k(aP) = (3,2) + (1,7)$
$= \underline{(5,3)}$

$Q_1 = kP = 13(1,4) = (3,2)$

Bob sends $((3,2), (5,3))$

Alice finds $aQ_1 = 11(3,2)$
$= (1,7)$

8 $\quad Q_2 - aQ_1 = (5,3) - (1,7)$

$$= (5,3) + (1,-7)$$

$$= \underline{(3,2)}.$$

This uses Diffie Hellman key <u>exchange protocol</u>.

$\cancel{\#(E)}$

$$\underbrace{q+1-2\sqrt{q}} \leq \# E \leq q+1+2\sqrt{q}.$$

$$\left| \# E - (q+1) \right| \leq 2\sqrt{q}.$$

<u>Hasse's bound for $\#(E)$</u>

$\mathbb{F}_p$ $\qquad$ $p \leq 11$

$\mathbb{F}_4, \mathbb{F}_9, \mathbb{F}_{25}$ — $\mathbb{F}_{p^2}$ $\qquad$ $p \leq 5$

$\mathbb{F}_8$ — $\mathbb{F}_{p^3}$ $\qquad$ $p \leq 2.$